

Literature Review

Monte Carlo Simulation of Blackjack Strategy, Expected Value and House Advantage

ACM 40960 • Monte Carlo Simulation

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Abstract

This literature review looks at the background needed for a project on simulating single-deck Blackjack using Monte Carlo methods. The project is mainly about estimating what happens when the game is played many times under a fixed set of rules. In particular, the review focuses on expected value, house advantage, player strategy, finite-deck probabilities and the uncertainty in simulation results. The main works discussed include the early mathematical study of Blackjack by Baldwin, Cantey, Maisel and McDermott, Thorp's work on card counting, later exact-probability approaches, and more recent computational and reinforcement-learning examples. Some mathematics is included because the project depends on clearly defined quantities such as payoff, expected return and confidence intervals. The review concludes that the best scope for this project is a clear single-deck simulator that compares a few simple strategies before adding optional extensions such as multiple decks, doubling, splitting or reinforcement learning.

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1 Introduction

Blackjack is a good project choice because it is easy to understand but still has enough mathematics to make the modelling interesting. The basic game is familiar: the player and dealer try to get close to 21 without going over, the player chooses whether to take more cards, and the dealer follows fixed rules. This makes the project more controlled than topics that need large datasets or complicated data cleaning. For this project, the data are produced by the simulation itself: a shuffled deck, a sequence of hands and the final outcome of each hand.

The ACM 40960 project list suggests starting with single-deck Blackjack and only expanding to two or four decks if the first version is manageable [1]. That is a sensible scope. A single-deck version is small enough to implement carefully, but it still includes the important feature that cards are removed from the deck after they are drawn. This means the probability of the next card changes as the hand progresses. That one detail is what separates finite-deck Blackjack from a much simpler model where every draw is treated as independent.

The aim of this literature review is to build a useful background for a Monte Carlo Blackjack simulator. The review is not trying to cover every possible gambling strategy. Instead, it focuses on the ideas that are most relevant to a student project: how Blackjack strategy became a mathematical problem, how expected value and house advantage are defined, how Monte Carlo simulation estimates these quantities, and how the results should be checked. The main research question is:

How can Monte Carlo simulation be used to estimate the expected value and house advantage of single-deck Blackjack under different player strategies?

The rest of the review follows the same general structure as the sample literature review document: it introduces the foundations, develops the modelling ideas, brings in the mathematical and computational literature, and then finishes by identifying the project gap and proposed direction.

2 Foundations of Blackjack as a Mathematical Problem

Blackjack became an important subject for probability because it is not a pure game of chance. Randomness decides which cards appear, but the player still has choices. For example, a player can hit, stand, double down or split depending on the rules being used. The value of each choice depends on the player's hand, the dealer's visible card and, in a finite-deck game, the cards that have already been removed.

The first major academic treatment was Baldwin, Cantey, Maisel and McDermott's 1956 paper, *The Optimum Strategy in Blackjack* [2]. Its importance is that it treated Blackjack decisions through expected value rather than instinct. In simple terms, each possible action can be assigned an average long-run payoff, and the best action is the one with the highest expected payoff. This idea is the basis of what is now called basic strategy.

Thorp then extended the mathematical study of the game by showing that the remaining deck composition matters [3, 4]. His work made card counting famous, but the more general modelling point is broader than counting cards in a casino. In a finite deck, the game has memory. If several ten-value cards have already been used, the chance of drawing another ten-value card is lower. If many low cards have been used, the remaining deck becomes richer in high cards. A

good single-deck simulation should capture this effect instead of drawing from a fixed probability distribution each time.

Later authors developed more exact methods for computing Blackjack probabilities and strategies. Solowiej, for example, discusses exact probability calculations for arbitrary deck compositions and compares these with approximate and Monte Carlo approaches [6]. This kind of work is more advanced than what is needed for the core project, but it is useful because it shows what a fully mathematical solution would look like. The project can use Monte Carlo simulation as the practical method while still using exact probability ideas for validation.

3 Mathematical Formulation for the Simulation

A simulation report is stronger when it states exactly what is being estimated. In this project, the central quantity is the expected payoff per hand. Before defining that, it is useful to describe the card probabilities in a single deck.

Let the ranks be

$$R = \{A, 2, 3, 4, 5, 6, 7, 8, 9, 10\},$$

where the rank 10 represents tens, jacks, queens and kings. At the start of a single deck,

$$N_A = 4, \quad N_2 = \dots = N_9 = 4, \quad N_{10} = 16.$$

If k cards have already been seen and c_r cards of rank r have appeared, then the probability that the next card has rank r is

$$\mathbb{P}(X_{k+1} = r \mid c_A, c_2, \dots, c_{10}) = \frac{N_r - c_r}{52 - k}. \quad (1)$$

This is the basic finite-deck equation. It is simple, but it is also the reason single-deck Blackjack is more realistic than an infinite-deck toy model.

A useful check for the simulator is the probability of a natural Blackjack on the first two player cards. This requires one ace and one ten-value card, in either order:

$$\mathbb{P}(\text{natural}) = 2 \left(\frac{4}{52} \right) \left(\frac{16}{51} \right) = \frac{128}{2652} \approx 0.0483. \quad (2)$$

So, in a large simulation, about 4.83% of initial two-card player hands should be naturals before taking the dealer's hand into account.

The payoff from a hand can be written as a random variable Y . A simple version is

$$Y = \begin{cases} +1.5, & \text{player has a natural and dealer does not,} \\ +1, & \text{player wins a normal hand,} \\ 0, & \text{push,} \\ -1, & \text{player loses.} \end{cases}$$

For a player strategy π and a ruleset $\mathcal{R}$, the expected value is

$$\mu_{\pi, \mathcal{R}} = \mathbb{E}[Y \mid \pi, \mathcal{R}]. \quad (3)$$

The house advantage is usually the negative of the player's expected value:

$$\text{HE}_{\pi, \mathcal{R}} = -\mu_{\pi, \mathcal{R}}. \quad (4)$$

For example, if the simulation estimates $\hat{\mu} = -0.006$, the implied house edge is about 0.6% of the initial stake.

Monte Carlo simulation estimates μ by repeating the game many times. If $Y_1, Y_2, \dots, Y_n$ are the simulated payoffs, then

$$\hat{\mu}_n = \frac{1}{n} \sum_{i=1}^n Y_i. \quad (5)$$

The sample variance and standard error are

$$\hat{\sigma}^2 = \frac{1}{n-1} \sum_{i=1}^n (Y_i - \hat{\mu}_n)^2, \quad \text{SE}(\hat{\mu}_n) = \frac{\hat{\sigma}}{\sqrt{n}}. \quad (6)$$

A simple 95% confidence interval is therefore

$$\hat{\mu}_n \pm 1.96 \frac{\hat{\sigma}}{\sqrt{n}}. \quad (7)$$

This matters because two strategies may appear different in a small simulation even when the difference is just random noise. Reporting a confidence interval makes the results more honest.

4 Monte Carlo Simulation and Strategy Evaluation

Monte Carlo simulation is useful here because the game can be repeated many times under controlled assumptions. The simulator can run 10^4 , 10^5 or 10^6 hands and record how often the player wins, loses, pushes, busts or gets a natural Blackjack. The main output should not just be the win rate. Expected value is more important because it includes the size of the payoff, especially when Blackjack pays 3:2 or when doubling is later introduced.

A simple implementation can compare three policies. The first is a naive policy, such as hitting until 17 and then standing. The second is a dealer-like policy, where the player behaves almost like the dealer. The third is a basic-strategy policy, where actions are chosen from a strategy table. The difference between two strategies can be written as

$$\Delta(\pi_1, \pi_2) = \hat{\mu}_{\pi_1, \mathcal{R}} - \hat{\mu}_{\pi_2, \mathcal{R}}. \quad (8)$$

This gives a clear way to report whether basic strategy performs better than a simpler rule.

Sutton and Barto use Blackjack as a standard example in reinforcement learning because it is an episodic decision problem [7]. An episode starts with the initial deal and ends with a win, loss or draw. The return from many episodes can be averaged to estimate the value of states and actions. This is closely related to Monte Carlo simulation, although reinforcement learning usually focuses on learning a policy rather than only evaluating one.

Modern software environments also use Blackjack as a testing ground. RLCard is a toolkit for reinforcement learning in card games and includes Blackjack among other games [8]. These tools are helpful for understanding how to organise a simulator into an environment and an agent. However, for this project, it is better to write a custom single-deck simulator because many

educational environments simplify the game, for example by drawing cards with replacement or using a limited action set.

The most important practical point from the literature is that simulation results must be validated. Equation (2) can check the opening deal. Dealer bust rates can be compared with published or analytically computed values under the same assumptions. Re-running the simulation with a larger n should reduce the standard error roughly like $1/\sqrt{n}$. These checks make the project look like a statistical model rather than only a programming exercise.

5 Basic Strategy, Card Counting and Rule Assumptions

Basic strategy is the natural benchmark for this project. It is not a guarantee of winning money, but it is the strategy that usually gives the best long-run result for a fixed set of casino rules when the player is not using detailed deck-composition information. It comes from the same expected-value idea introduced in the early Blackjack literature [2].

Card counting is more advanced, but it helps explain why a single-deck simulation is interesting. In the Hi-Lo count, low cards 2–6 receive value +1, cards 7–9 receive value 0, and tens and aces receive value −1. If $x_1, \dots, x_k$ are the exposed cards and $w(x_j)$ is the counting weight, the running count is

$$R_k = \sum_{j=1}^k w(x_j). \quad (9)$$

A positive or negative count gives a rough signal about the remaining deck. Thorp’s work showed that this kind of information can change the player’s expectation [3, 4]. In a coursework project, card counting does not need to be the main focus, but a simple count-based extension would be a good optional addition after the basic simulator is working.

Rule assumptions are also very important. Blackjack is not one single game in every source. Results change depending on whether the dealer stands or hits on soft 17, whether Blackjack pays 3:2 or 6:5, whether doubling and splitting are allowed, how many decks are used, and how deep the deck is dealt before shuffling. Recent work on simplified Blackjack variants also shows that changing the rules can change both basic strategy and expectation [9]. Bordeu and Castro’s dynamic-programming study similarly treats the strategy and betting problem as rule-dependent [10].

For this reason, the final project should define one baseline ruleset clearly. A good first version is:

- one standard 52-card deck;
- dealer stands on all 17s;
- natural Blackjack pays 3:2;
- no surrender;
- start with hit and stand only, then add double or split if time allows.

Each later rule change can then be treated as a separate experiment. If $\mathcal{R}_1$ and $\mathcal{R}_2$ are two different rulesets, the effect of changing rules can be measured by

$$\Delta_{\mathcal{R}_1, \mathcal{R}_2}(\pi) = \mu_{\pi, \mathcal{R}_1} - \mu_{\pi, \mathcal{R}_2}. \quad (10)$$

This is a simple but useful way to make the project more mathematical without making it too complicated.

6 Synthesis

The literature points toward a clear project design. The early mathematical work explains why Blackjack strategy can be studied through expected value. Thorp's work explains why the remaining deck matters. Exact-probability studies show that a fully analytical solution is possible, but also more complex than needed for the main project. Monte Carlo simulation gives a practical middle ground: it is understandable, flexible and easy to extend.

For this project, the strongest approach is to build a finite single-deck simulator from scratch. The simulator should begin with a simple ruleset and a small number of player policies. It should then report expected value, house edge, win rate, bust rate, natural frequency, standard error and a confidence interval. These outputs connect directly to the mathematics in Section 3 and to the literature on strategy evaluation.

A useful final results table could look like this:

Metric	Definition	Purpose
Win rate	n_{win}/n	Easy summary of outcomes
Expected value	$n^{-1} \sum_i Y_i$	Main measure of strategy quality
House edge	$-\hat{\mu}$	Casino-style interpretation
Standard error	$\hat{\sigma}/\sqrt{n}$	Precision of the estimate
95% interval	$\hat{\mu} \pm 1.96SE$	Shows uncertainty
Natural frequency	n_{BJ}/n	Checks opening deal logic

The main gap this project fills is not that Blackjack has never been studied before. It clearly has. The value of the project is that it brings the main ideas together in a manageable simulation: finite-deck probability, simple strategy comparison and statistical uncertainty. That is enough for a strong modelling project without turning it into a full professional Blackjack engine.

7 Conclusion

Blackjack is a suitable topic for ACM 40960 because it connects probability, simulation and decision-making in a compact setting. The rules are simple enough to implement, but the outcomes are complex enough to justify a careful literature review and a mathematical model. The literature shows a clear progression from basic expected-value strategy, to finite-deck card counting, to exact probability methods and modern computational approaches.

The recommended project scope is a single-deck Monte Carlo simulator with clearly stated rules. The simulator should compare a few strategies and report results with confidence intervals rather than only percentages. If time allows, the project can be extended to multiple decks, doubling, splitting, a simple card-counting rule or a reinforcement-learning comparison. However, the core work should remain simple, transparent and well validated. In that form, the project is realistic to complete and still has enough mathematical content to be academically meaningful.

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